

PROBLEM STATEMENT #47 | Developer Tools & IT Operations**Domain & SSL Certificate Expiry Alert System****1. Problem Context & Overview**

An IT operations utility that performs automated daily WHOIS registration and SSL/TLS handshake audits, alerting sysadmins through escalation ladders before domain or certificate expiration.

Target Stakeholders / Actors: *SysAdmin, Security Officer*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.

Sample Acceptance Criteria: Pass: Alert queued for certificate expiring in 15 days; Fail: Expired certificate ignored without notification.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.